Remedy

Healthcare RAG Agent

Framework: LangChain + LangGraph + Streamlit | Version: 1.0 | Date: June 2026
Repository: bibhu2020/agenticai (Hugging Face Space)

1. Overview

Remedy is a healthcare information retrieval assistant powered by Retrieval-Augmented Generation (RAG). It first searches a local FAISS vector store built from healthcare documents. If the RAG result is insufficient, it automatically falls back to a live DuckDuckGo web search. The application is built on LangChain + LangGraph and served via Streamlit.

2. System Design

2.1 Components

Component	Technology	Purpose
app.py	Streamlit	Chat UI with session history
chat.py	Python helper	Formats message history and calls the graph
healthcare_agent.py	LangGraph StateGraph	Defines the agent graph: agent node ↔ tools node
rag.py	FAISS + HuggingFace Embeddings	Builds and queries the local vector store
tools/rag_tool.py	LangChain @tool	RAG search tool exposed to the agent
DuckDuckGoSearchRun	langchain_community	Web search fallback tool
llm.py	LangChain BaseChatModel	LLM factory (openai | google | groq)

2.2 Architecture Diagram

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User (Streamlit chat)
■
chat.py ■■■■ builds HumanMessage list
■
LangGraph StateGraph
■■■■■■■▼■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ START → agent_node ■
■ ■■■ bind_tools([rag_search, ddg_search]) ■
■ ■ ■
■ ▼ tools_condition ■
■ tools_node ■
■ ■■■ rag_search → FAISS vector store ■
■ ■■■ duckduckgo_search → live web ■
■ ■ ■
■ ■■■ back to agent_node ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ final AIMessage
Streamlit renders markdown response
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3. Agentic Flow

Step	Actor	Action
1	agent_node (LLM)	Receive user question + SystemMessage; decide which tool to call
2	rag_search (tool)	Query FAISS with question embedding; return top-k chunks
3	agent_node	Evaluate RAG result: useful? → synthesise; not useful? → call DuckDuckGo
4	duckduckgo_search (tool)	Live web search for real-time or missing information
5	agent_node	Synthesise all tool results into structured Markdown healthcare response
6	Streamlit	Append response to session chat history and display

4. RAG Setup

- Vector store: FAISS (Facebook AI Similarity Search) — fast in-memory similarity search
- Embeddings: HuggingFace sentence-transformers (all-MiniLM-L6-v2)
- Documents: Healthcare PDFs / text files loaded from a configurable data directory
- Index: Built at startup if not already cached; stored locally on disk
- Retrieval: Top-3 most similar chunks returned per query

5. Configuration

Env Var	Required for
OPENAI_API_KEY	Default LLM (gpt-4o-mini)
GOOGLE_API_KEY	Gemini model option
GROQ_API_KEY	Groq model option